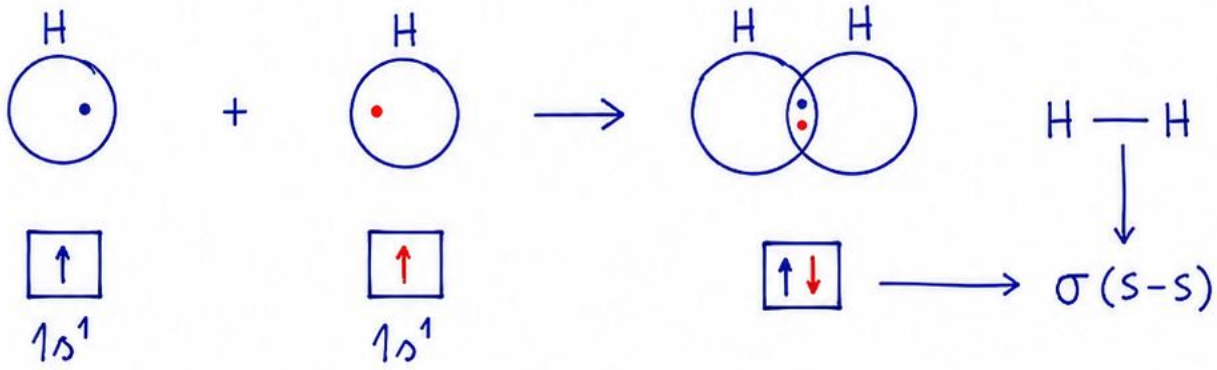
HIBRIDIZAÇÃO

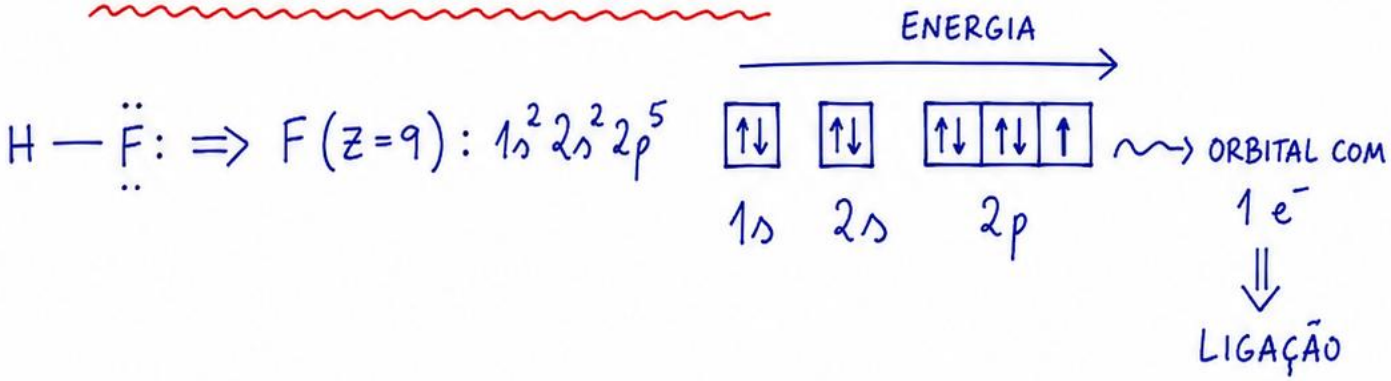
VSEPR ⟶ GEOMETRIAS ⟵ ORBITAIS

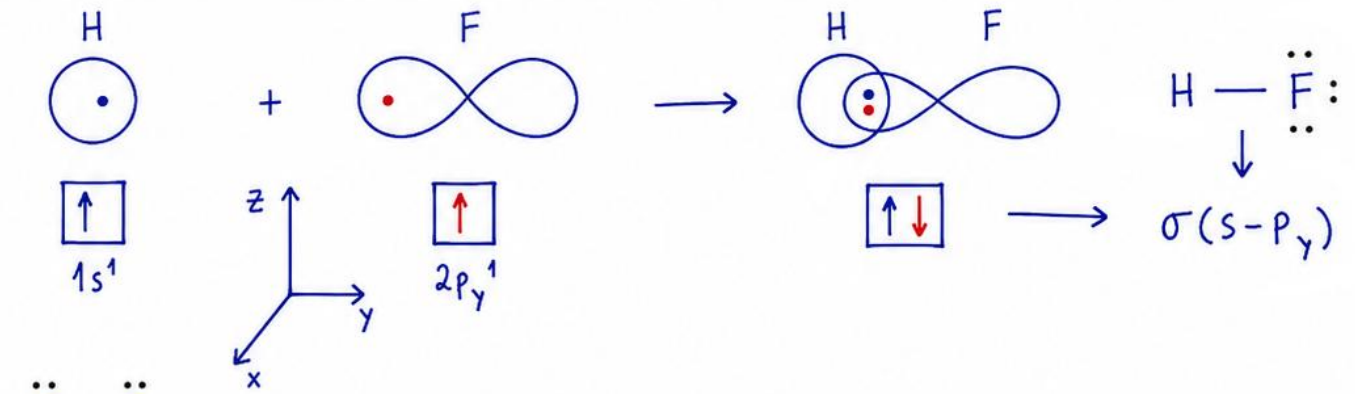
- HIBRIDIZAÇÃO : EXPLICAR LIGAÇÃO QUÍMICA A PARTIR DOS ORBITAIS DOS ÁTOMOS

✳ LIGAÇÕES SIMPLES, DOIS ÁTOMOS



- LIGAÇÃO QUÍMICA É A SOBREPOSIÇÃO DE ORBITAIS ATÔMICOS COM 1 e⁻
- LIGAÇÃO SIGMA (σ): FORMADA PELA SOBREPOSIÇÃO DOS ORBITAIS AO LONGO DO EIXO DA LIGAÇÃO



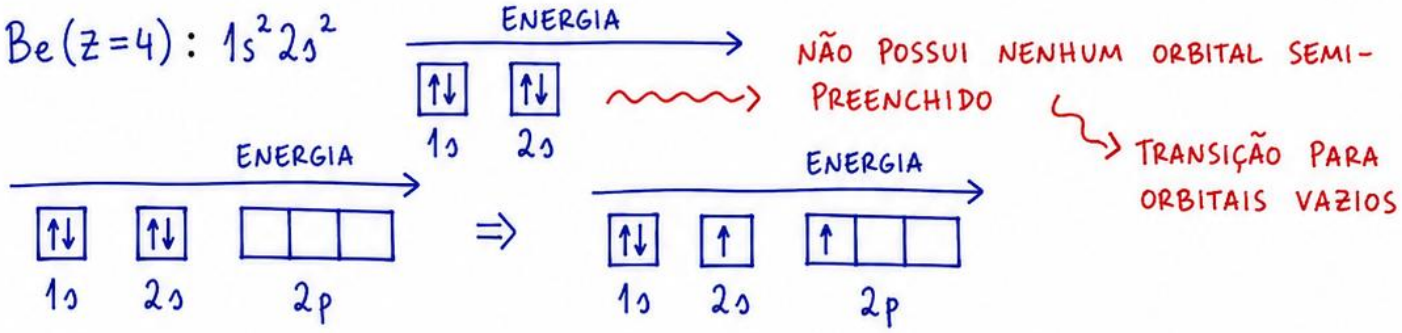


✳ LIGAÇÕES SIMPLES, MAIS ÁTOMOS

BeH_2 ⇒ H — Be — H ⇒ H — Be — H (180°) ⇒ 2 DOMÍNIOS LIGANTES ⇓ PRECISA DE 2 ORBITAIS SEMI-PREENCHIDOS

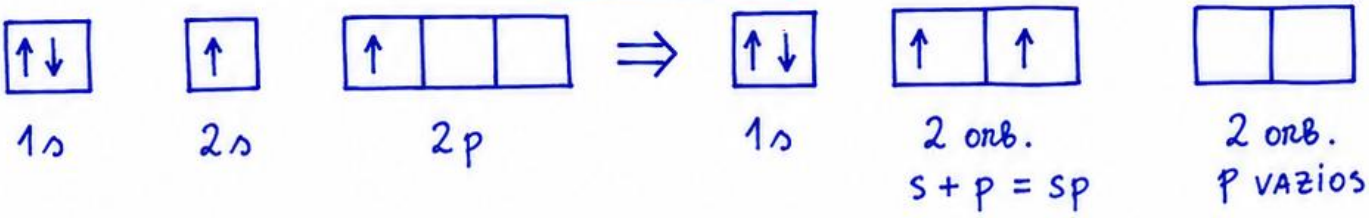
FÓRMULA MOLECULAR — ESTRUTURA DE LEWIS — GEOMETRIA MOLECULAR

$Be\ (Z=4): 1s^2 2s^2$

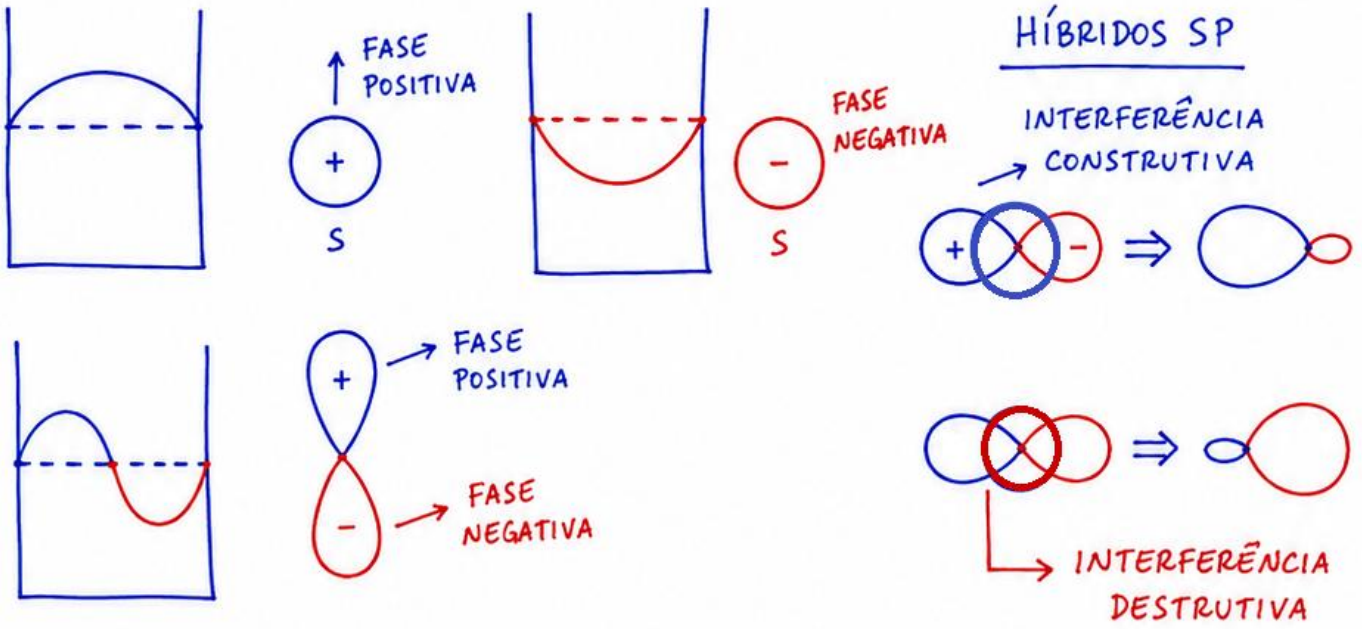


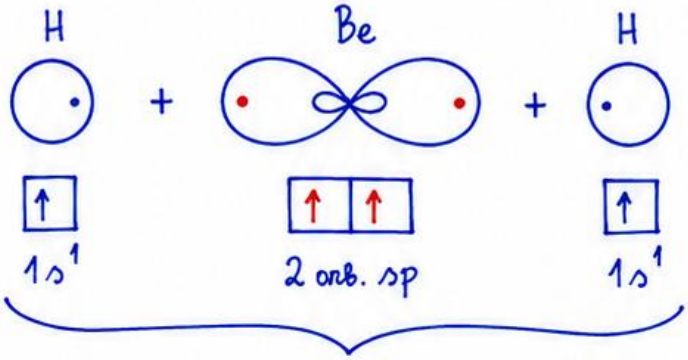
- PARA FORMAR LIGAÇÕES EQUIVALENTES, DEPOIS DA TRANSIÇÃO É PRECISO COMBINAR OS ORBITAIS S E P COM $1e^-$ PARA RESULTAR EM UM MESMO NÚMERO DE ORBITAIS HÍBRIDOS.

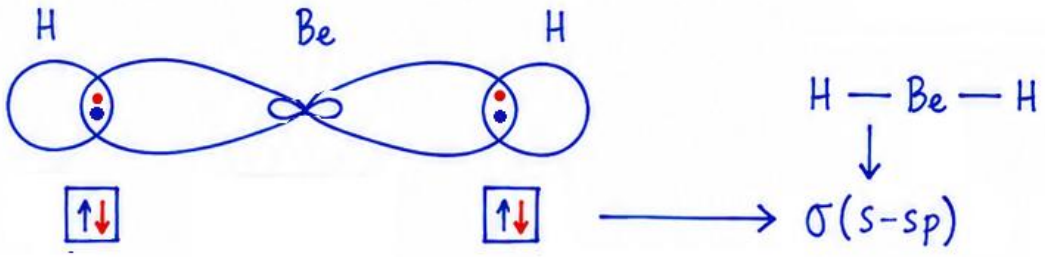
HIBRIDIZAÇÃO

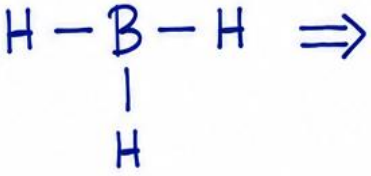


- OS ORBITAIS HÍBRIDOS SP RESULTAM DA INTERFERÊNCIA CONSTRUTIVA E DESTRUTIVA DOS ORBITAIS S E P





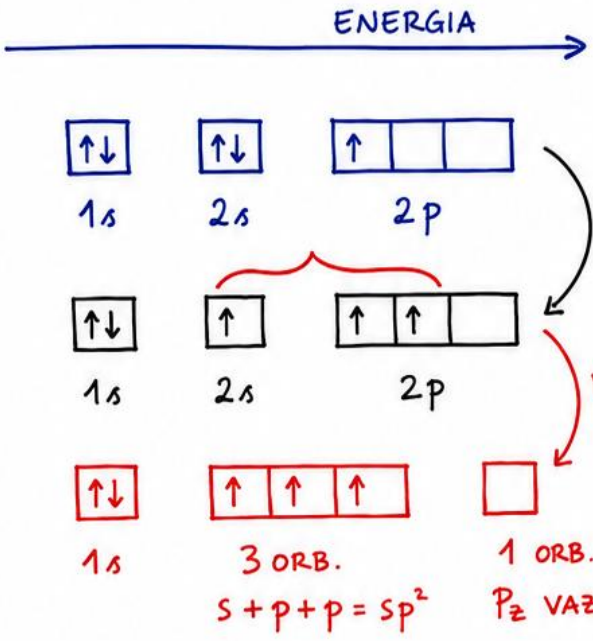


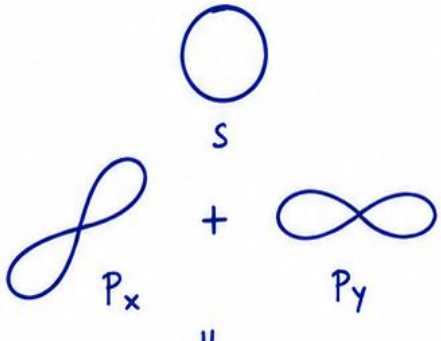
$BH_3 \Rightarrow$  $\Rightarrow$  $\Rightarrow$ 3 DOMÍNIOS LIGANTES

$\Downarrow$

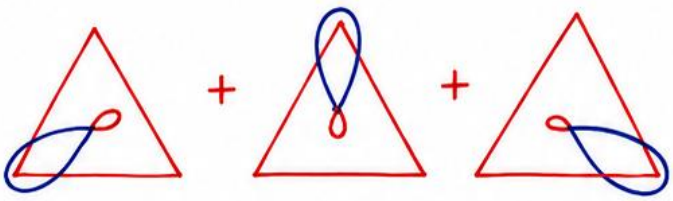
DEVE TER 3 ORBITAIS SEMI-PREENCHIDOS EQUIVALENTES (MESMO FORMATO)

$B\,(z=5): 1s^2\,2s^2\,2p^1 \rightarrow$ APENAS 1 ORBITAL SEMI-PREENCHIDO

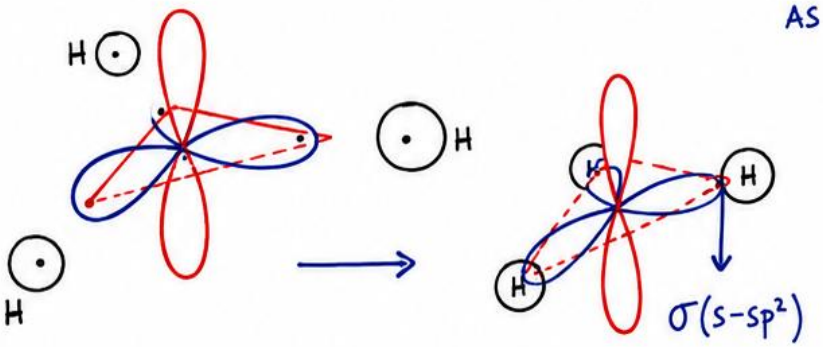


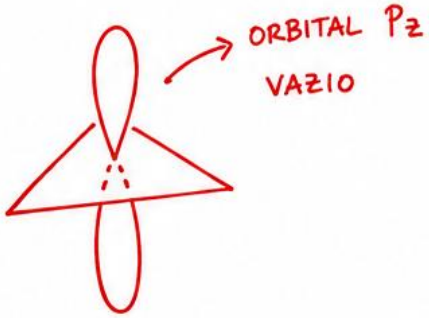


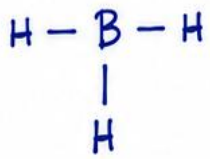
$\Downarrow$

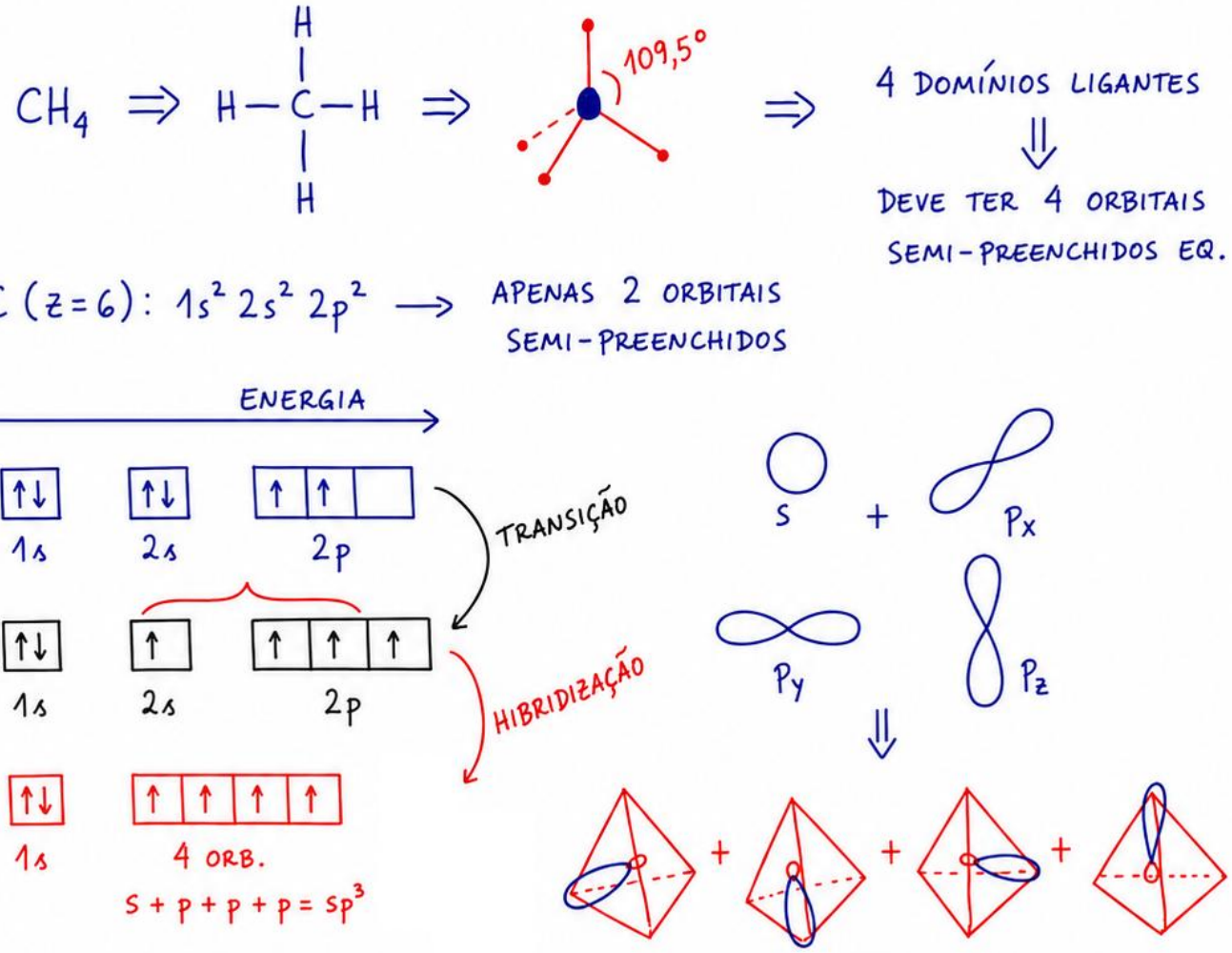


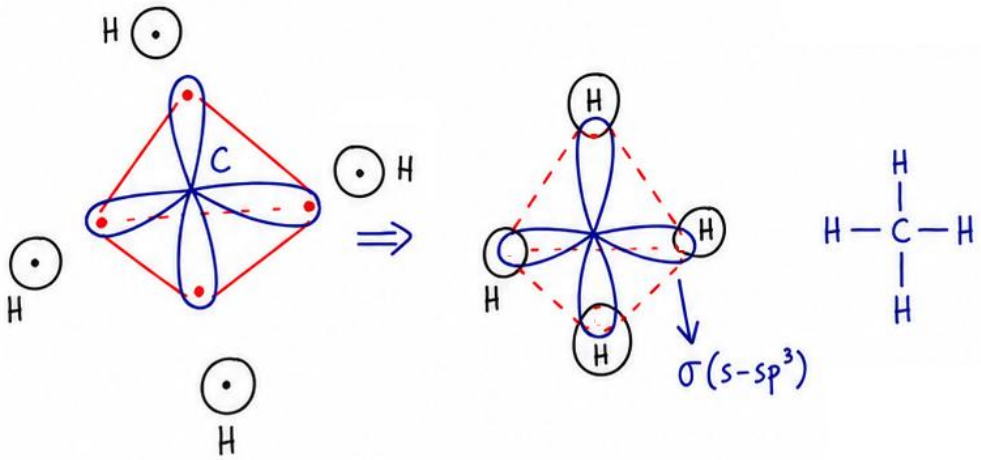
ORBITAIS HÍBRIDOS APONTAM PARA AS ESTREMIDADES DE UM TRIÂNGULO





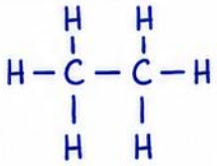
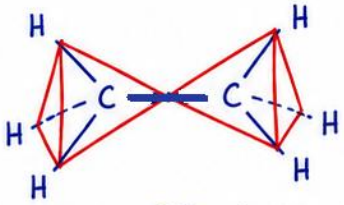


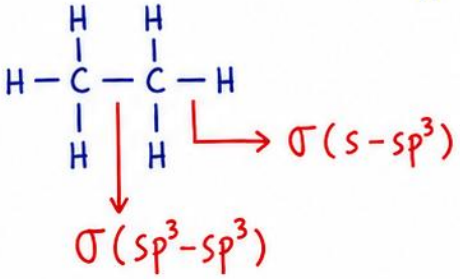


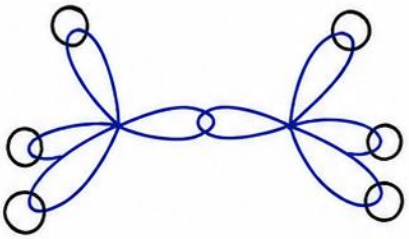


DL	HIBRIDIZAÇÃO
1	NENHUMA (S OU P)
2	sp
3	sp^2
4	sp^3
5	sp^3d
6	sp^3d^2

✳ LIGAÇÕES MÚLTIPLAS

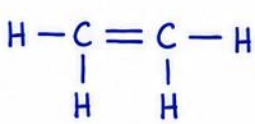
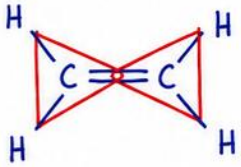
- ETANO (C_2H_6) ⇒  ⇒ 

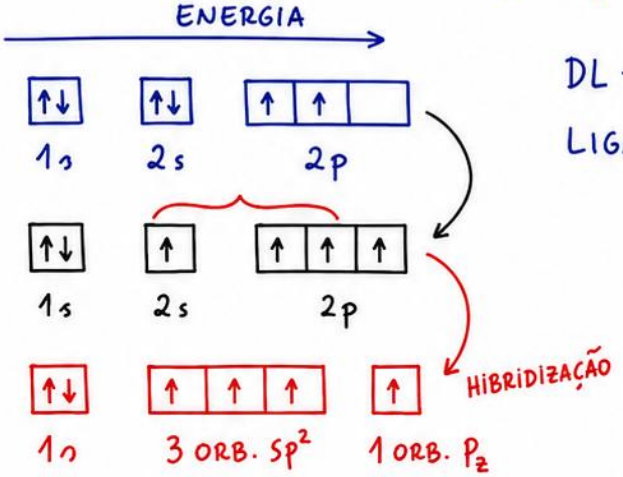




DE = 4, DL = 4 ⇒ HIBRIDIZAÇÃO

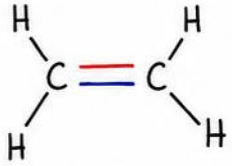
S + P + P + P = sp^3

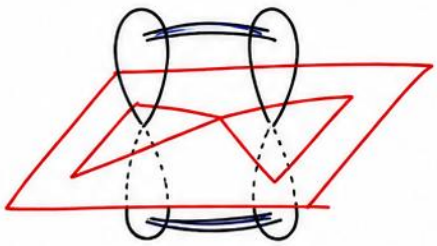
- ETENO (C_2H_4) ⇒  ⇒ 



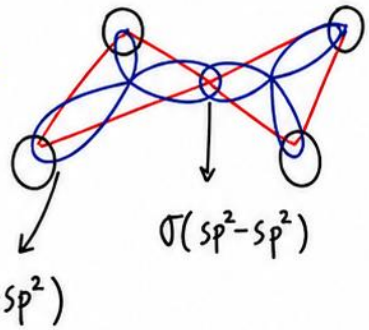
DL → HIBRIDIZAÇÃO → σ

LIGAÇÕES MÚLTIPLAS → ORBITAIS P NÃO HIBRIDIZADOS COM 1 ELÉTRON

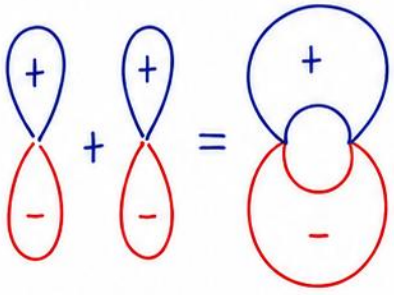




SIGMA:

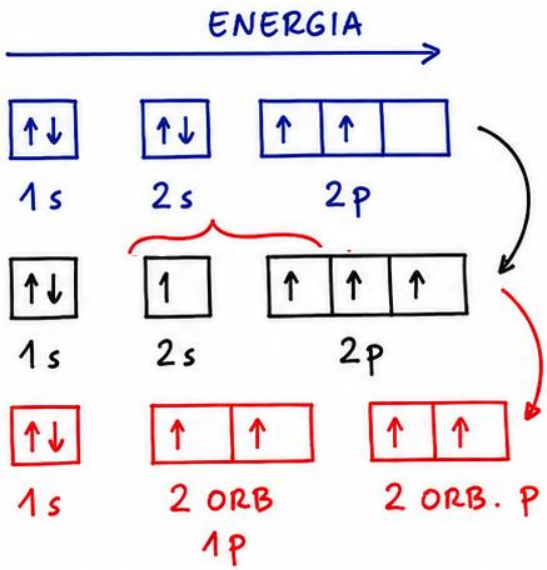


PI (π): SOBREPOSIÇÃO LATERAL (PERPENDICULAR AO EIXO DA LIGAÇÃO) DOS ORBITAIS p

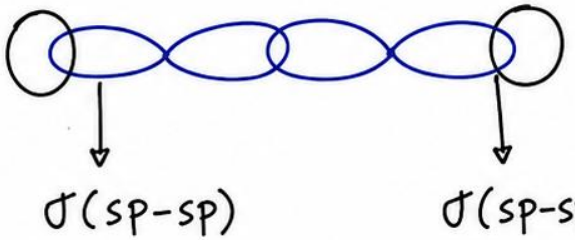


- ETINO (C_2H_2) $\Rightarrow$ $H-C\equiv C-H$ $\Rightarrow$ —H—C≡C—H— (180°)

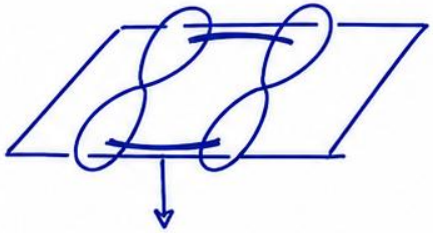
DE = 2, DL = 2 → HIB. SP

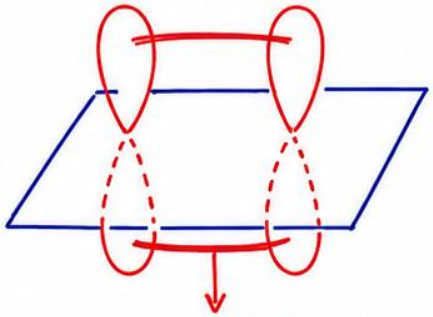


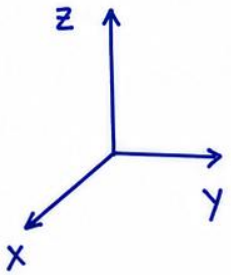
SIGMA :

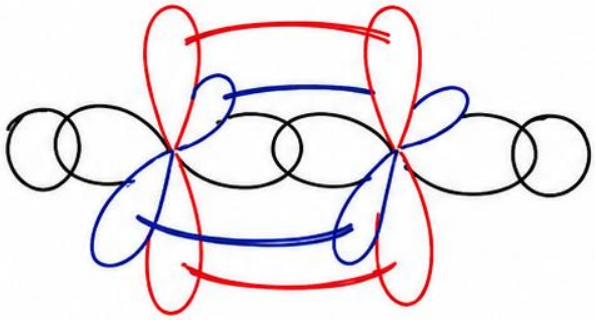


PI :



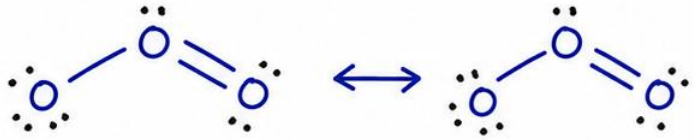


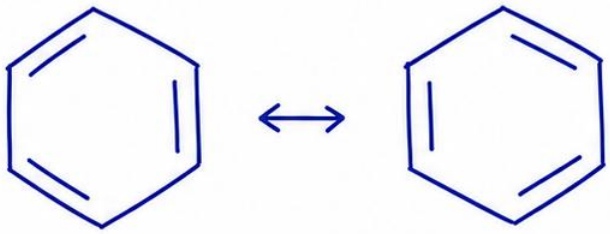




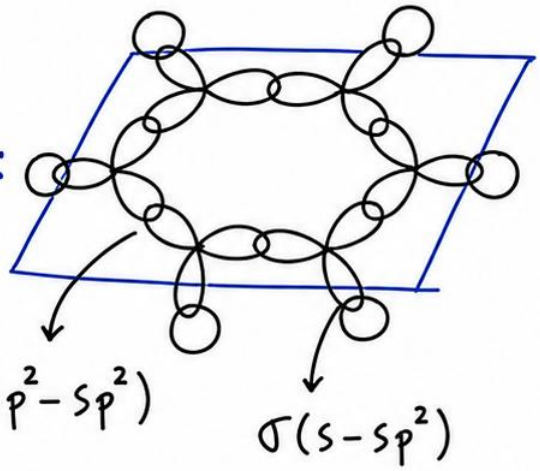
NÃO EXISTE TRIPLA LIGAÇÃO π POIS EIXOS (E ORBITAIS p) ACABAM

✳ RESSONÂNCIA

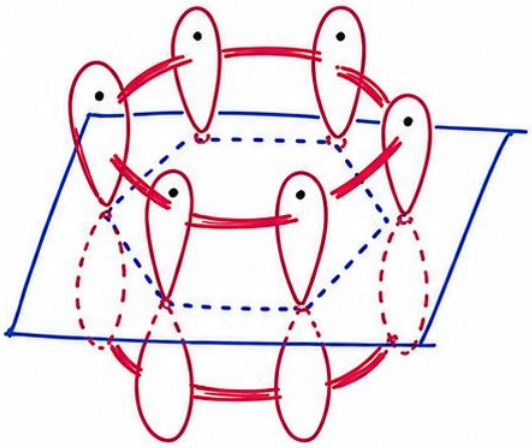




SIGMA:



PI:



A RESSONÂNCIA SE DEVE A SOBREPOSIÇÃO DE ORBITAIS P COM UM ELÉTRON QUANDO SE TEM LIGAÇÕES MÚLTIPLAS ALTERNADAS